2 Types of studies and Plots

2.1 Plotting the data

We will start with a **scatter plot**.

A scatter plot is a tool for examining the relationship between two numerical variables X and Y. It plots each observed (x,y) pair as a dot on the x-y plane.

```
?DNase  #what is the dataset about?
head(DNase) #shows the first rows
```

```
  Run       conc density
1   1 0.04882812   0.017
2   1 0.04882812   0.018
3   1 0.19531250   0.121
4   1 0.19531250   0.124
5   1 0.39062500   0.206
6   1 0.39062500   0.215
```

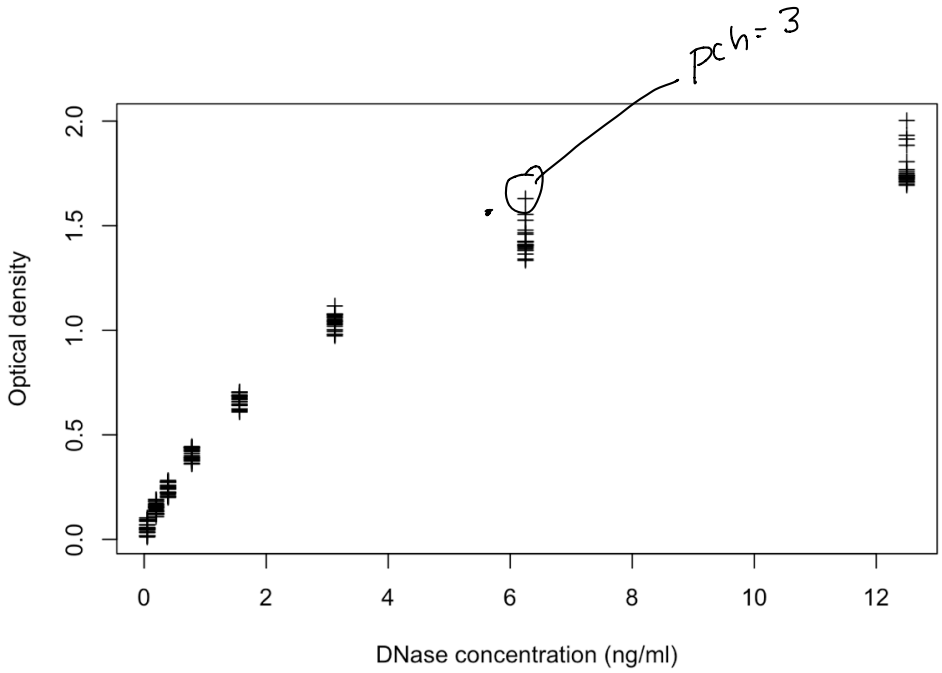
```
names(DNase) # gives you the name of the variables
```

```
[1] "Run"     "conc"    "density"
```

X Y

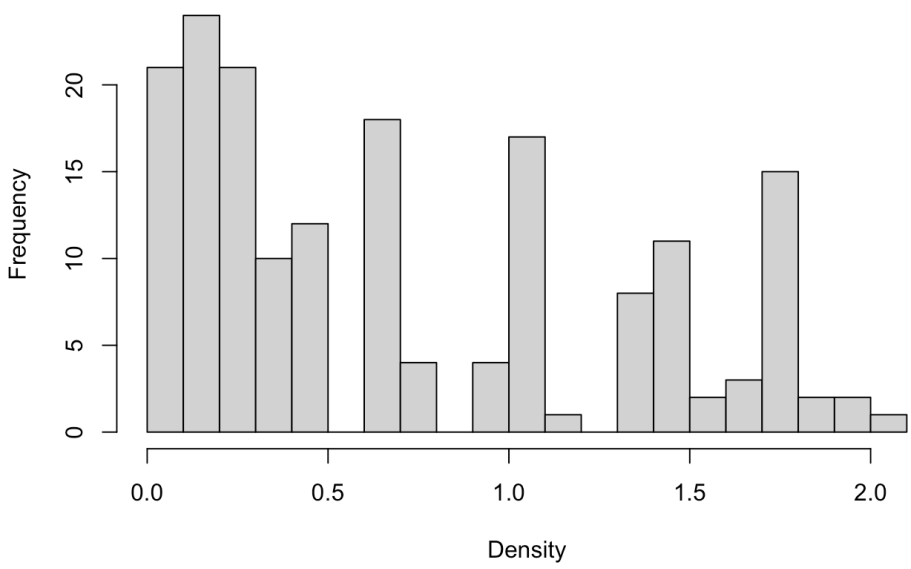
```
plot(DNase$conc, DNase$density,
  ylab = attr(DNase, "labels")$y,
  xlab = paste(attr(DNase, "labels")$x, attr(DNase, "units")$x),
  pch = 3)
```

mark

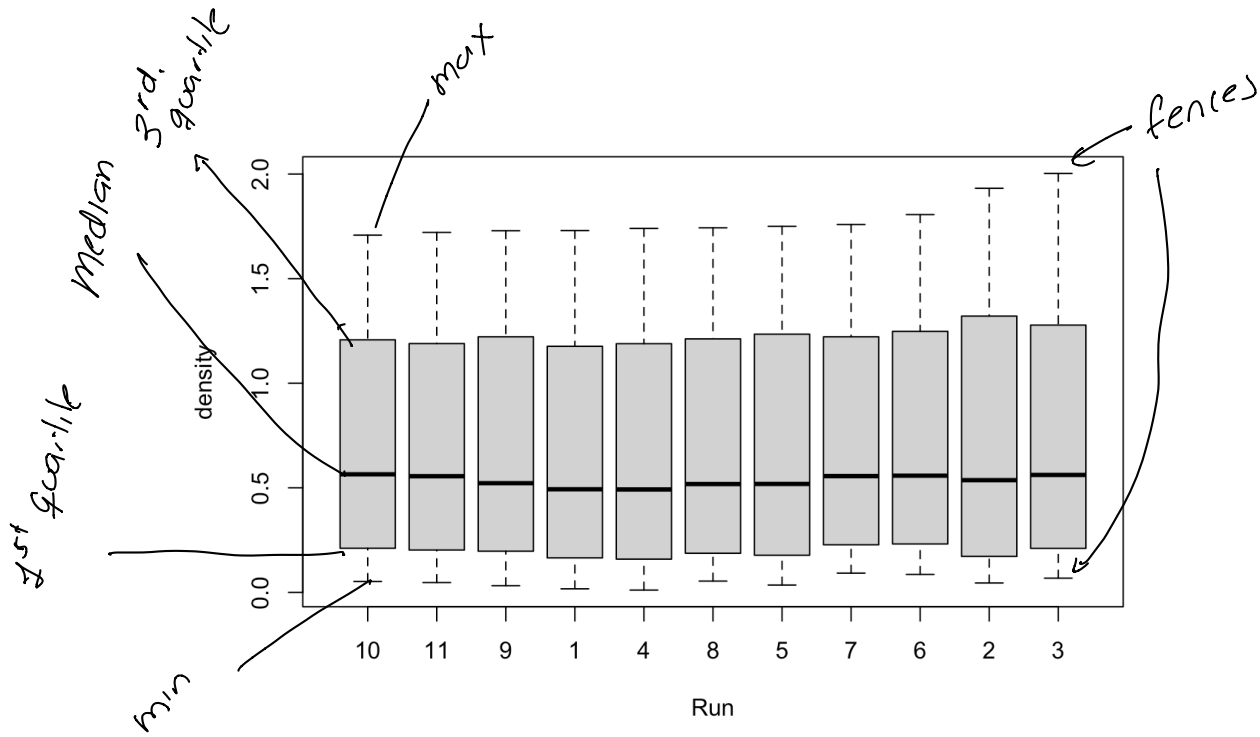


We can visualize *frequency distributions* with **bar plots** (as well as jittered dotplots), **histograms**, and **boxplots**. Bar plots and histograms display the frequency of values (or grouped values). A boxplot displays the minimum, 1st quartile, median, 3rd quartile, and maximum at once. We can also compare multiple distributions at once.

```
hist(DNase$density, breaks=25, main = "",xlab="Density")
```




```
boxplot(density ~ Run, data = DNase)
```



The **shape** of a distribution is often informative. It may be, for example: (a) symmetric and bell-shaped, (b) symmetric but not bell-shaped, (c) right-skewed, (d) left-skewed, (e) exponential, or (f) bimodal.

Base R plotting functions are good for quick exploration, but they have aesthetic limitations. We will use a framework called the **grammar of graphics**, implemented in the package **ggplot2**.

2.2 Types of Studies

Researchers gather information to make inferences about nature in a variety of settings. Much of statistics deals with **data analysis**, but statistical considerations often play a key role in the planning and design of scientific investigation.

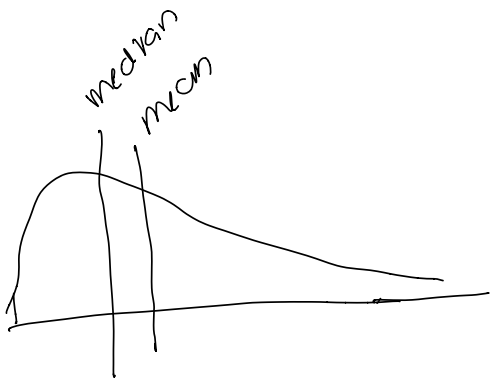
Data in biology are obtained from either an **experimental study** or an **observational study**. In an experimental study, the researcher designs how to randomly assign different treatments to the individual units of study. In an observational study, on the other hand, the researcher has no control over which units fall into which groups.

Example: Marriage and Life expectancy

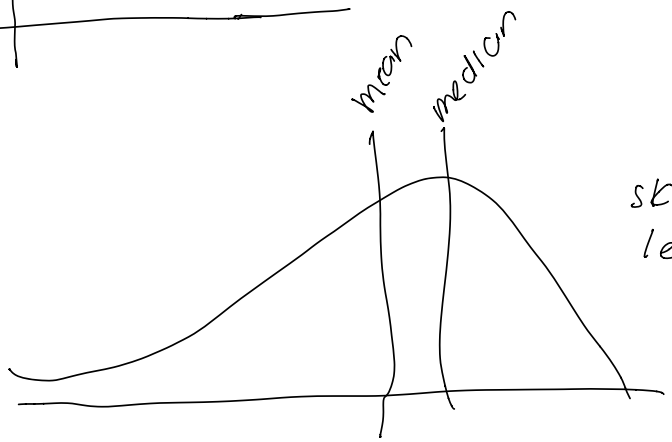
A study found that people who were married are more likely to live longer than single. However, from an **observational** study such as this we don't know whether marriage increases life expectancy or whether persons who are healthier are more likely to get married.

Example: Aids as confounder

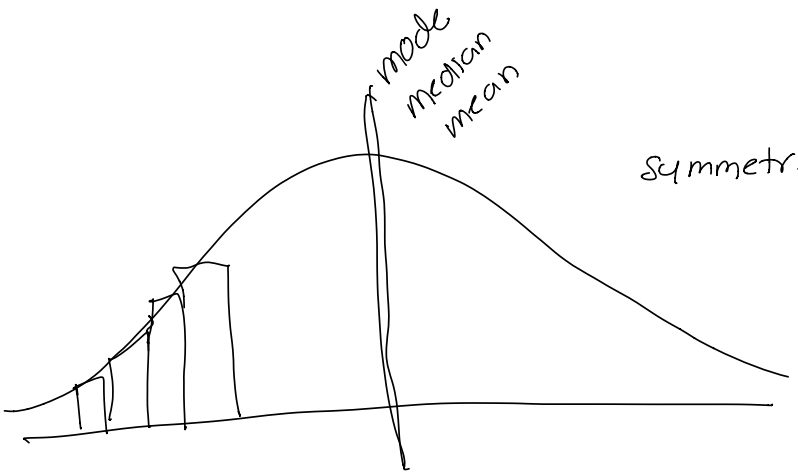
Some research has suggested that there is a biological basis for sexual orientation. One such study involved measuring a section of the brain for 40 homosexual men, 40



Skewed to the right



skewed to the left.

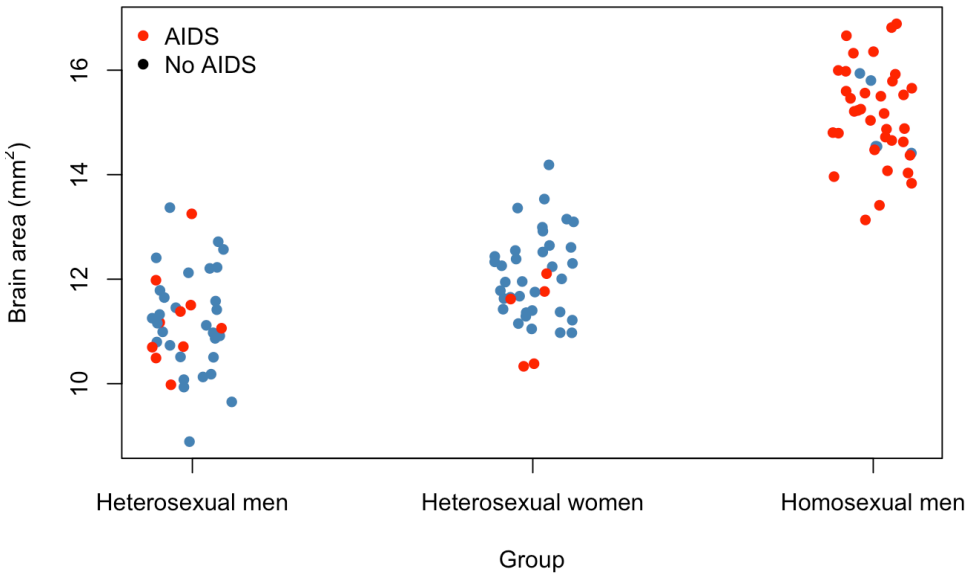


symmetric

heterosexual men, and 40 heterosexual women. The research found that the brain section tends to be larger in homosexual men and heterosexual women than in heterosexual men.

Average brain area (mm^2) by group

group	Brain Area
Heterosexual men	11.19
Heterosexual women	12.01
Homosexual men	15.15



The previous two examples are based on **observational studies**. In an observational study the researcher systematically collects data from subjects, but only as an observer and not as someone who is controling the conditions. However, observational studies can be misleading due to **confounding variables**. The context in which the data arose is important and tell us to be on the alert for the effects that other factors, such as the impact of AIDS, may have on the size of the section of the brain.

Confounders are variables, other than the treatment, which are responsible for causing (at least part of) the observed difference in the outcome between the various levels of the treatment.

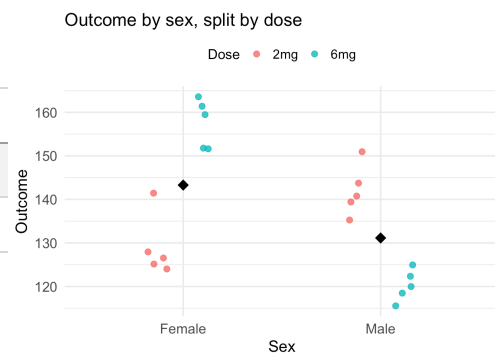
How can we mitigate confounding? We can design the experiment and create treatment and control groups that are as similar as possible except for the treatment level. We can anticipate and prevent other known and unknown explanations for the observed difference in outcomes between the treated and control groups.

Example: Effect of drug on cats

In part of one study, a new drug for lowering cholesterol was given to seven male and seven female cats at doses of $2mg/kg$ and $6mg/kg$. The two doses were assigned at random to the seven cats within each sex and cholesterol levels were measured after some time.

Summary by Sex and Dose

Sex	Mean (2mg)	Mean (6mg)
Female	129.0132	157.5753
Male	142.0265	120.2499



The **design of this experiment** allows for the investigation of the **interaction** between two factors: sex of the cat and dose. That is, for females, the effect of increasing the dose was positive in lowering cholesterol, but for males the effect of increasing the dose was negative.

By **randomly assigning** treatments (drug doses) to subjects (cats), we can get around the problem of confounding that complicates observational studies and limits the conclusions. **Randomized controlled experiments** are considered the gold standard in scientific investigation, but they can also be difficult.

Often, human subjects in experiments are given a **placebo**. Of course, if a placebo control is used, then the subjects must not be told which group they are in, that is, the experiment is **blinded**. The purpose of blinding is to minimize the extent to which subject expectations influence the results of the experiment. If subjects exhibit a psychological reaction to getting a medication, that placebo response will tend to balance out between the two groups, so that any difference between the groups can be attributed to the effect of the active treatment.

Placebo

Example: Peanut sensitivity

A group of people who reported mild peanut sensitivity took part in a study. They were given a flavored drink and were told it contained a small amount of peanut extract and might trigger itching, hives, or throat tightness. After drinking it, several participants reported tingling in the mouth, skin itching, and mild breathing discomfort.

Later, the same participants were given an “anti-allergy” drink that they were told would block reactions. This time, most reported little or no symptoms.

In reality, both drinks were identical and contained no peanut. The difference in symptoms appeared to be driven by expectation/power of suggestion, with negative expectations producing symptoms (nocebo) and reassuring expectations reducing them (placebo).

How about the person evaluating the responses? This person can be subject to bias influencing the observation process itself. An experiment in which both the subjects and the persons making the evaluations of the response are blinded is called **double-blinded**.

Example: “Adherence effect” in a blood-pressure trial (why you need a control group)

Researchers test a new blood-pressure pill intended to reduce heart attacks. As the study goes on, some participants reliably take their pills, while others often miss doses. After 4 years, the researchers split the treatment group into high adherers (took ≥80% of pills) and low adherers (<80%).

At first glance, results look impressive: high adherers in the treatment group have far fewer heart attacks than low adherers—so it might seem the drug works.

But the study also included a placebo control group. When researchers do the same adherence split in the placebo group, they see the same pattern: placebo participants who took ≥80% of their pills also had fewer heart attacks than placebo participants who didn’t.

Illustrative example: apparent benefit among adherers appears in both drug and placebo groups (adherer effect).

Group	Adherence	N	4-year event rate (%)
New BP drug	≥ 80% pills taken	500	4.0
New BP drug	< 80% pills taken	500	9.0
Placebo (control)	≥ 80% pills taken	500	4.5
Placebo (control)	< 80% pills taken	500	9.5

Interpretation: the lower event rate is largely an "adherer effect"—people who consistently follow the protocol may also exercise more, eat better, keep doctor appointments, and manage other risks. Without the placebo control group, the researchers could mistakenly credit the drug for benefits that are actually due to differences between adherers and nonadherers.

Randomized allocation in medical settings may not be ethical. One possibility is to use **historical controls** (i.e. previous patients with the same illnes who were treated with another therapy). The challenge is that overall characteristics of the patients may change with time. A possible solution is to do some statistical adjustment to control for differences in characteristics (for example age).

2.3 Sampling

Random Sampling

In order to answer scientific questions, we first must consider the scope of our study and identify our population. Typically we cannot observe the entire population and obtain a sample. How we gather our data should inform us about our choice of analysis method and on the validity of the study. Ideally our sample will be a **representative** subset of the population and avoid obtaining a **biased** sample.

In a **simple random sample** (SRS) of size n, (a) every member of the population has the same chance of being included in the sample, and (b) each member is chosen independently of each other.

```
set.seed(123)
names<-seq(1,100) #data frame
sample(names,5,replace=FALSE)
```

```
[1] 31 79 51 14 67
```

What woud be some practical concerns about SRS?

Two other common sampling techniques are **the random cluster sample** where the sampling unit is the cluster and the **stratified random sample** where the sampling unit is the individual within a strata.

Most of the examples in this class will assume SRS.

2.4 Sampling error

How can we use a sample summary (statistic) to **infer** a population value? The answer is: with some error!. We will see later how statistical theory enables us to set limits on the amount of error. In particular, **sampling error** since we only have a sample. If the sample is not a simple random sample (SRS), it may be *biased*: some individuals are more likely to be selected than others.

2.5 Nonsampling Errors

For example, the type of variable we are collecting may affect how truthfuly people respond. The way in which questions are worded may also influence the respond. **Nonresponse bias** resulting in bias caused by persons not responding to some of the questions in a survey or not returning the questionnaire. In general, patterns of missing data (MCAR, MAR, or MNAR) hamper studies by introducing bias.

- MCAR (Missing Completely At Random): The probability a value is missing does not depend on any data (neither observed nor unobserved).
- MAR (Missing At Random): The probability a value is missing may depend on observed data, but not on the missing value itself after conditioning on what's observed.

-MNAR (Missing Not At Random): The probability a value is missing depends on the unobserved value itself (or other unobserved factors), even after accounting for observed data.

It is then best to make every effort to avoid missing data. But if data are missing, it is crucial that the possible reasons are considered when interpreting and reporting results.